Deepak Kandel

(+1) 585–410–8148 dk6012@rit.edu Rochester, NY, USA https://github.com/kandeldeepak46

Summary

Machine learning researcher with hands-on experience in image processing, computer vision, and deep learning from research & development to production. Currently, I am experimenting with developing novel algorithms to estimate uncertainty, incorporate self-awareness, and support continual/lifelong learning for streaming data to address real-world problems (vision, language) using probabilistic inference and foundation models.

Education

Doctor of Philosophy (Ph.D.), Imaging Science **Aug 2022 — Present**

Chester F. Carlson Center For Imaging Science, Rochester Institute of Technology Rochester, New York, U.S.A.

- **Courses**: Image Processing & Computer Vision, Fourier Methods, Human Vision, Deep Learning, Optimization

Bachelor's Degree, Electronics & Communication Engineering **Nov 2015 — Sep 2019**

DoECE, Pulchowk Campus, Institute of Engineering, Tribhuvan University Lalitpur, Bagmati, Nepal

- Merit-based 4-year scholarship
- **Courses**: Artificial Intelligence, Calculus, Linear Algebra, Probability & Statistics, Digital Signal Processing

Research Interests

Bayesian Optimization, Parameter-Efficient Fine-Tuning (PEFT) of Large Language Models (LLMs), Uncertainty Quantification and Calibration of LLMs, Lifelong / Continual Learning.

Work Experience

Graduate Research Assistant **June 2023 — Present**

Rochester Institute of Technology Rochester, New York, U.S.A.

- Experimenting with variational inference and pre-trained-models (ViT) for lifelong/continual learning
- Developing algorithms for incremental object detection (IOD) using YOLO and vision-language models (VLM).
- Uncertainty-aware fine-tuning of Large Language Model (LLM)for robust and reliable generalization.

Machine Learning Engineer (Internship) **May 2025 — Aug 2025**

MRIMath Inc. Birmingham, AL, U.S.A.

- Medical Image Segmentation: Metastatic Brain Tumor, Organ at Risk (OAR), and Head & Neck Cancer)
- Parameter-Efficient Fine-Tuning (PEFT) with an Adapter of LLM for Brain Tumor Segmentation

Graduate Teaching Assistant **Aug 2022 — May 2023**

Rochester Institute of Technology, Rochester, New York, U.S.A.

- GTA For Probability & Statistics with Python
- GTA For Fundamentals of Imaging Science

Machine Learning Engineer **Dec 2020 — July 2022**

Evolve Kathmandu, Bagmati, Nepal

- Designed and developed a Multi–Store–Multi–Item demand quantity prediction system using ARIMA, Prophet, Temporal Fusion Transformer, and LightGBM, and deployed in AWS
- Developed the product recommendation engine using Autoencoders and LightFM for **Procter & Gamble, Britannia, Kellogg's, Mars**, and **Unilever**'s FMCG products in Nepal
- Developed in-house Outlet Localization (OL) and Automatic Route Generator (ARC) REST APIs based on geospatial data for optimized distribution of goods

Data Scientist **Mar 2020 — Dec 2020**

Docsumo Kathmandu, Bagmati, Nepal

- Information retrieval from tax, invoice, and bill documents and digitization of non-digital PDF, bank statements, W2, and W9 forms using advanced OCR techniques
- Contributed to a face recognition system for the verification of documents with an Indian driving license
- Improving the image quality — skewness detection, pixel correction, and enhancements with traditional filtering techniques and computer vision algorithms
- Developed and deployed the AI applications in Google Cloud Platform

Projects

- Training and Fine-Tuning Large Language Models (LlaMa-2) for NeurIPS Large Language Model Efficiency Challenge
- Assessment of U-Net for Brain Tumor Segmentation with Gaussian Noise on Training Data
- Book Recommendation System based on Hybrid Methods of Matrix Filtering
- Instance Segmentation of Buildings using Deep CNN For Paint Color Visualization
- Near-Real Time Objection Recognition using Pre-trained Deep Neural Network
- Sentiment Analysis of Google Play applications reviews with Fine-Tuning of BERT
- Digital Image Halftoning and Removal Using Fast Fourier Transform
- MRI Image Denoising using Markov Chain Monte Carlo and Non-Local Means

Technical Stacks

Programming	python, c/c++
Deep Learning / Computer Vision	PyTorch
LLMs & GenAI	LoRA Adaptation, Visual Prompt Tuning
Machine Learning	scikit-learn, NumPy, pandas, matplotlib, SciPy
Image Processing	OpenCV
Project Management	Gitlab, Jira, Confluence
Miscellaneous	Cloud Computing, Git, Bash, Flask, Docker, MLOps

Awards & Honors

Scholarship/Assistantship for Ph.D.

- Received merit-based scholarship at RIT to pursue Ph.D. in Imaging Science

Full Scholarship for Undergraduate Studies

- Government-funded 4-year Scholarship For Engineering Studies (4% acceptance rate)

Training's & Crash Course

Data Analytics & Machine Intelligence (DAMI) **Jan 2019**

- 15-day workshop organized by ICTC Center, Institute of Engineering (IoE), Pulchowk Campus, in association with IIT Delhi

Kickstart for AI & Machine Learning **Jun 2018**

- Took part in the workshop by AI Dev Nepal to get a kickstart on the basics and AI and Machine Learning

Workshop on Set Farm **Jan 2018**

- Workshop on set farm Nepal for intelligence in farming & agriculture with robotics, organized by Purwanchal Campus Dharan, Pulchowk Campus, Nepal, & Washington State University, USA